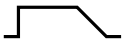
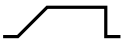
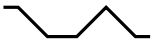
Math Kangaroo Practice

Grade 3-4 · 24 questions · harder · seed 66554

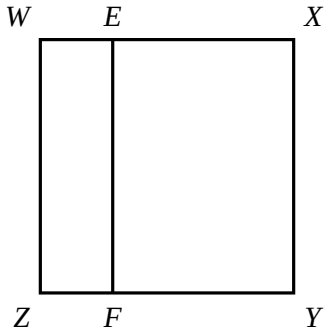
Part A: Each question is worth 3 points

1. This is the outline of a town seen from the river at sunset.

Which piece **cannot** be part of the outline?

A. ***B.***  ***C.***  ***D.***  ***E.***

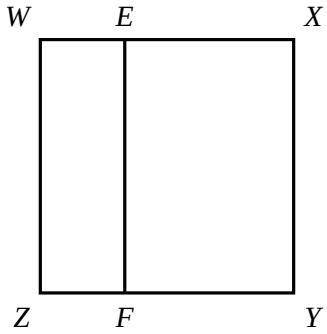
2. *WXYZ* is a square. Its side is 21 cm. The line *EF* cuts it into two rectangles, and *WE* = 6 cm.



How many centimetres is the perimeter of the square *WXYZ* larger than that of the rectangle *EXYF*?

A. 24 cm ***B.*** 12 cm ***C.*** 10 cm ***D.*** 9 cm ***E.*** 6 cm

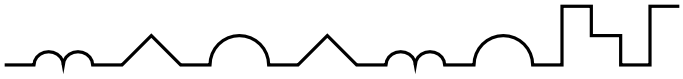
3. *WXYZ* is a square. Its side is 18 cm. *WEFZ* is a rectangle. Its shorter side is 6 cm.



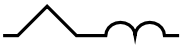
How many centimetres is the perimeter of the square *WXYZ* larger than that of the rectangle *WEFZ*?

A. 12 cm ***B.*** 23 cm ***C.*** 24 cm ***D.*** 36 cm ***E.*** 48 cm

4. Lena cut a strip of paper along this line.

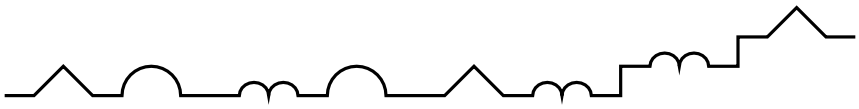


Which piece **cannot** come from the cut line?

A. ***B.*** ***C.***  ***D.*** ***E.***

Part B: Each question is worth 4 points

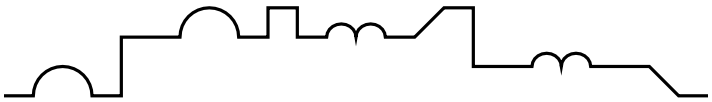
5. Lena cut a strip of paper along this line.



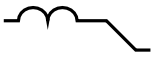
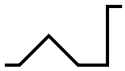
Which piece **can** come from the cut line?

A. ***B.*** ***C.*** ***D.*** ***E.***

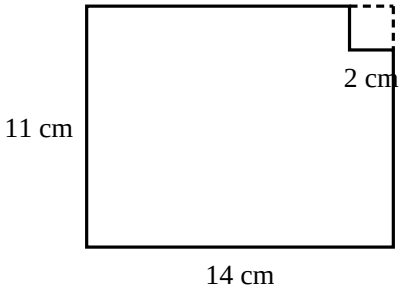
6. This is the outline of a town seen from the river at sunset.



Which piece **can** be part of the outline?

A. ***B.***  ***C.*** ***D.*** ***E.*** 

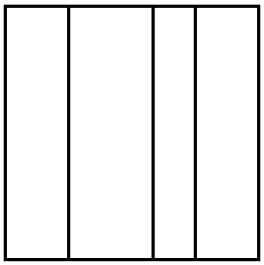
7. A rectangle is 14 cm long and 11 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 50 cm ***B.*** 48 cm ***C.*** 46 cm ***D.*** 42 cm ***E.*** 25 cm

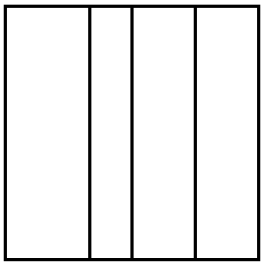
8. Grandma bakes a square cake. A ribbon 52 cm long fits exactly around it. She cuts the cake into 4 rectangular pieces as shown and puts a ribbon around every piece.



How many centimetres of ribbon does she need for all the pieces together?

A. 208 cm ***B.*** 156 cm ***C.*** 131 cm ***D.*** 130 cm ***E.*** 91 cm

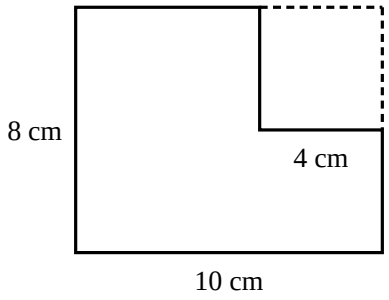
9. A square has perimeter 80 cm. It is cut along straight lines into 4 rectangles as shown.



What is the sum of the perimeters of all 4 rectangles?

A. 80 cm ***B.*** 120 cm ***C.*** 140 cm ***D.*** 160 cm ***E.*** 200 cm

10. A rectangle is 10 cm long and 8 cm wide. A square with side 4 cm is cut off one corner.

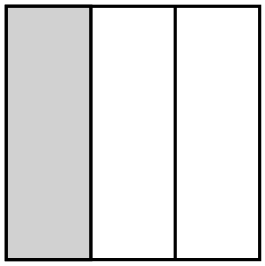


What is the perimeter of the remaining shape?

A. 20 cm ***B.*** 28 cm ***C.*** 32 cm ***D.*** 36 cm ***E.*** 52 cm

Part C: Each question is worth 5 points

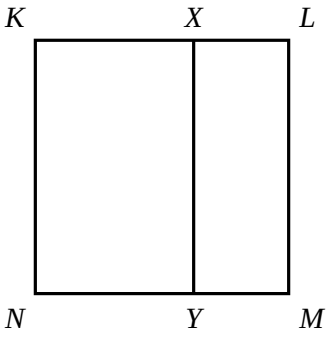
11. Mia cuts a square sheet of paper into 3 equal strips as shown. The perimeter of each strip is 56 cm.



What is the perimeter of the square?

A. 168 cm ***B.*** 112 cm ***C.*** 104 cm ***D.*** 101 cm ***E.*** 84 cm

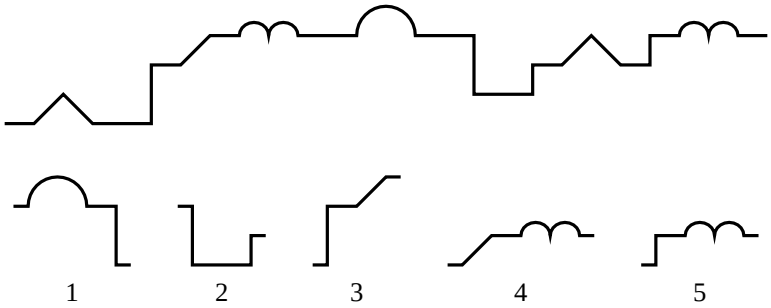
12. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 52 cm and the perimeter of *XLMY* is 44 cm.



What is the perimeter of the square *KLMN*?

A. 64 cm ***B.*** 66 cm ***C.*** 75 cm ***D.*** 80 cm ***E.*** 96 cm

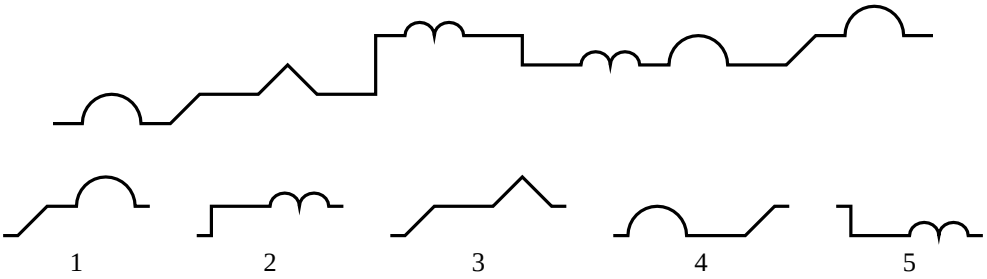
13. This is the outline of a town seen from the river at sunset.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

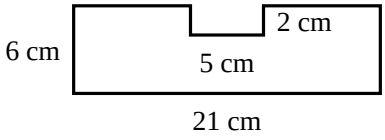
14. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 5

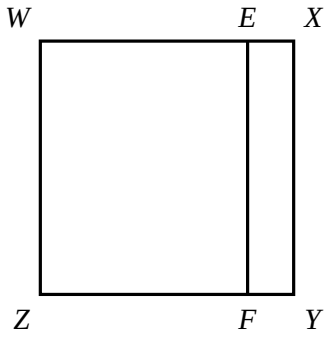
15. A 21 cm by 6 cm rectangle of cardboard has a rectangular notch 5 cm wide and 2 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 49 cm ***B.*** 56 cm ***C.*** 58 cm ***D.*** 64 cm ***E.*** 68 cm

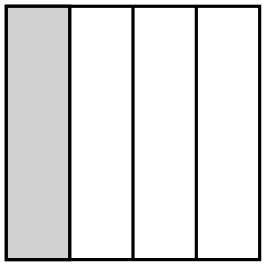
16. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 40 cm and the perimeter of *EXYF* is 26 cm.



How long is the side of the square *WXYZ*?

A. 16 cm ***B.*** 11 cm ***C.*** 10 cm ***D.*** 9 cm ***E.*** 8 cm

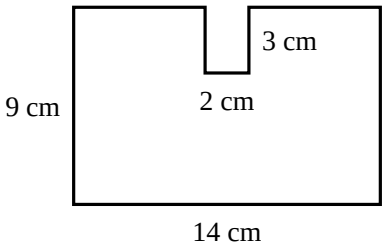
17. Mia cuts a square chocolate bar into 4 equal strips as shown. The perimeter of each strip is 50 cm.



What is the perimeter of the square?

A. 200 cm ***B.*** 100 cm ***C.*** 96 cm ***D.*** 85 cm ***E.*** 80 cm

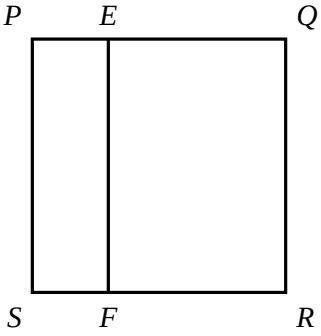
18. A 14 cm by 9 cm rectangle of cardboard has a rectangular notch 2 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 52 cm ***B.*** 50 cm ***C.*** 49 cm ***D.*** 46 cm ***E.*** 44 cm

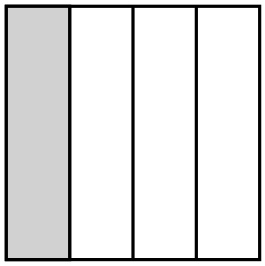
19. The square *PQRS* is cut into two rectangles. The perimeter of *PEFS* is 26 cm and the perimeter of *EQRF* is 34 cm.



What is the perimeter of the square *PQRS*?

A. 40 cm ***B.*** 30 cm ***C.*** 20 cm ***D.*** 15 cm ***E.*** 8 cm

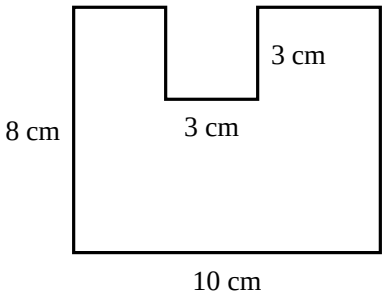
20. Mia cuts a square sheet of paper into 4 equal strips as shown. The perimeter of each strip is 50 cm.



How long is the side of the square?

A. 5 cm ***B.*** 10 cm ***C.*** 15 cm ***D.*** 18 cm ***E.*** 20 cm

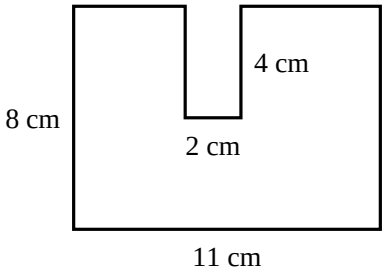
21. A 10 cm by 8 cm rectangle of cardboard has a rectangular notch 3 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 42 cm ***B.*** 43 cm ***C.*** 45 cm ***D.*** 47 cm ***E.*** 48 cm

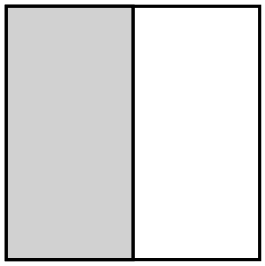
22. A 11 cm by 8 cm rectangle of cardboard has a rectangular notch 2 cm wide and 4 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 50 cm ***B.*** 46 cm ***C.*** 42 cm ***D.*** 38 cm ***E.*** 36 cm

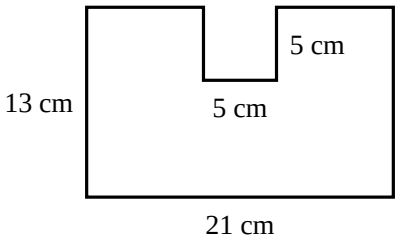
23. Mia cuts a square sheet of paper into 2 equal strips as shown. The perimeter of each strip is 24 cm.



What is the perimeter of the square?

A. 8 cm ***B.*** 16 cm ***C.*** 32 cm ***D.*** 40 cm ***E.*** 48 cm

24. A 21 cm by 13 cm rectangle of cardboard has a rectangular notch 5 cm wide and 5 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 78 cm ***B.*** 73 cm ***C.*** 68 cm ***D.*** 63 cm ***E.*** 60 cm

Answer Key

1.	**C**	**2.**	**B**	**3.**	**C**	**4.**	**A**	**5.**	**B**	**6.**	**C**
7.	**A**	**8.**	**D**	**9.**	**E**	**10.**	**D**	**11.**	**E**	**12.**	**A**
13.	**E**	**14.**	**D**	**15.**	**C**	**16.**	**B**	**17.**	**E**	**18.**	**A**
19.	**A**	**20.**	**E**	**21.**	**A**	**22.**	**B**	**23.**	**C**	**24.**	**A**

Solutions

1. C — All the other pieces can be found in the line, but this one cannot: it is a mirror image of a part of the line. [2002 Q7 · cannot_belong]

2. B (12 cm) — *EXYF* has sides 15 and 21 cm, perimeter 72 cm. Square: 84 cm. Difference 12 cm (= 2 × 6). [2002 Q6 · square_minus_strip]

3. C (24 cm) — Square: 4 × 18 = 72 cm. Rectangle: 6 + 18 + 6 + 18 = 48 cm. Difference 72 – 48 = 24 cm. [2002 Q6 · square_minus_strip]

4. A — All the other pieces can be found in the line, but this one cannot: it has a round dome where the line has two small bumps. [2002 Q7 · cannot_belong]

5. B — Only this piece is found in the line. The others don't fit — A: it is a mirror image of a part of the line; C: it steps down where the line steps up; D: it is a mirror image of a part of the line; E: its flat part between the two features has the wrong length. [2002 Q7 · can_belong]

6. C — Only this piece is found in the line. The others don't fit — A: it steps down where the line steps up; B: its flat part between the two features has the wrong length; D: it has the right shapes but in the wrong order; E: it has a pointed roof where the line has a round dome. [2002 Q7 · can_belong]

7. A (50 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (14 + 11) = 50 cm. [2002 Q6 · corner_cut]

8. D (130 cm) — Side = 52 ÷ 4 = 13 cm. There are 3 cuts; each cut is 13 cm long and is counted twice (once for each piece next to it): 52 + 3 × 2 × 13 = 130 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

9. E (200 cm) — Side = 80 ÷ 4 = 20 cm. There are 3 cuts; each cut is 20 cm long and is counted twice (once for each piece next to it): 80 + 3 × 2 × 20 = 200 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

10. D (36 cm) — The two new edges (4 cm and 4 cm) replace the two removed pieces of the outer edge, which were also 4 cm and 4 cm. So the perimeter is the same as the rectangle's: 2 × (10 + 8) = 36 cm. [2002 Q6 · corner_cut]

11. E (84 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 56 ÷ 8 = 7, so the side is 3 × 7 = 21 cm and the perimeter is 84 cm. [2002 Q6 · strips_to_square]

12. A (64 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 52 + 44 = 96 cm = 6 sides. Side = 96 ÷ 6 = 16 cm, perimeter 4 × 16 = 64 cm. [2002 Q6 · two_perimeters]

13. E (4) — Pieces 2, 3, 4, 5 fit. Not: 1 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]

14. D (4) — Pieces 1, 3, 4, 5 fit. Not: 2 (it is a mirror image of a part of the line). [2002 Q7 · how_many_fit]

15. C (58 cm) — The bottom of the notch (5 cm) replaces the 5 cm taken out of the top edge, and the two sides of the notch add 2 × 2 cm. Perimeter = 2 × (21 + 6) + 2 × 2 = 58 cm. [2002 Q6 · notch]

16. B (11 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 40 + 26 = 66 cm = 6 sides. Side = 66 ÷ 6 = 11 cm. [2002 Q6 · two_perimeters]

17. E (80 cm) — A strip is the full side long and 1/4 of it wide, so its perimeter is side + side + 2 × (side ÷ 4) = 10 × (side ÷ 4). 50 ÷ 10 = 5, so the side is 4 × 5 = 20 cm and the perimeter is 80 cm. [2002 Q6 · strips_to_square]

18. A (52 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (14 + 9) + 2 × 3 = 52 cm. [2002 Q6 · notch]

19. A (40 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 26 + 34 = 60 cm = 6 sides. Side = 60 ÷ 6 = 10 cm, perimeter 4 × 10 = 40 cm. [2002 Q6 · two_perimeters]

20. E (20 cm) — A strip is the full side long and 1/4 of it wide, so its perimeter is side + side + 2 × (side ÷ 4) = 10 × (side ÷ 4). 50 ÷ 10 = 5, so the side is 4 × 5 = 20 cm. [2002 Q6 · strips_to_square]

21. A (42 cm) — The bottom of the notch (3 cm) replaces the 3 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (10 + 8) + 2 × 3 = 42 cm. [2002 Q6 · notch]

22. B (46 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 4 cm. Perimeter = 2 × (11 + 8) + 2 × 4 = 46 cm. [2002 Q6 · notch]

23. C (32 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 24 ÷ 6 = 4, so the side is 2 × 4 = 8 cm and the perimeter is 32 cm. [2002 Q6 · strips_to_square]

24. A (78 cm) — The bottom of the notch (5 cm) replaces the 5 cm taken out of the top edge, and the two sides of the notch add 2 × 5 cm. Perimeter = 2 × (21 + 13) + 2 × 5 = 78 cm. [2002 Q6 · notch]